crop, that the human race is doomed within the lifetime of Cooper's grandchildren.

What Catastrophes?

What kind of catastrophes could have produced Cooper's world? Our biologist experts offered a number of possible, but improbable, answers. Here are several:

Leadbetter: Today (2008) most people aren't growing their own food. We're dependent on a global system for growing and distributing food, and for distributing water. You could imagine that system breaking down due to some biological or geophysical catastrophe. As an example on a small scale, if there was no snow in the Sierra Nevada Mountains for a few consecutive years, there would be little drinking water in Los Angeles. Ten million people would be forced to migrate, and agricultural output in California would plummet. You can easily imagine much larger scale catastrophes. In Cooper's world, with a vastly reduced population and a return to agrarian society, the production and distribution problems are lessened.

Simon: Another possible catastrophe: Over human history there has been a continual battle between us and *pathogens* (microbes that attack the human body or attack plants or other animals). We humans have developed a sophisticated immune system to deal with the pathogens that attack us directly. But the pathogens keep evolving and we're always half a step behind them. At some point there could be a catastrophe where the pathogens change so fast that our immune systems can't keep up.